



BOARD QUESTION PAPER : JULY 2023

MATHEMATICS AND STATISTICS

Time: 3 Hrs.

Max. Marks: 80

General instructions:

The question paper is divided into **FOUR** sections.

- (1) **Section A:** Q. 1 contains **Eight** multiple choice type of questions, each carrying **Two** marks.
Q. 2 contains **Four** very short answer type questions, each carrying **one** mark.
- (2) **Section B:** Q. 3 to Q. 14 contain **Twelve** short answer type questions, each carrying **Two** marks.
(Attempt any **Eight**)
- (3) **Section C:** Q. 15 to Q. 26 contain **Twelve** short answer type questions, each carrying **Three** marks.
(Attempt any **Eight**)
- (4) **Section D:** Q. 27 to Q. 34 contain **Eight** long answer type questions, each carrying **Four** marks.
(Attempt any **Five**)
- (5) Use of log table is allowed. Use of calculator is not allowed.
- (6) Figures to the right indicate full marks.
- (7) Use of graph paper is not necessary. Only rough sketch of graph is expected.
- (8) For each multiple choice type of question, it is mandatory to write the correct answer along with its alphabet, e.g., (a)..... / (b)..... / (c)..... / (d)....., etc. No marks shall be given, if ONLY the correct answer or the alphabet of correct answer is written. Only the first attempt will be considered for evaluation.
- (9) Start answer to each section on a new page.

SECTION – A

Q.1. Select and write the correct answer for the following multiple choice type of questions: [16]

- i. The dual of statement $p \wedge \sim q$ is equivalent to _____.
(a) $\sim p \wedge q$ (b) $p \leftrightarrow q$ (c) $\sim p \vee q$ (d) $\sim p \rightarrow \sim q$ (2)
- ii. If $|\vec{a}| = 3$, $|\vec{b}| = 4$, then a value of λ for which $\vec{a} + \lambda \vec{b}$ is perpendicular to $\vec{a} - \lambda \vec{b}$ is _____.
(a) $\frac{9}{16}$ (b) $\frac{3}{4}$ (c) $\frac{3}{2}$ (d) $\frac{4}{3}$ (2)
- iii. The acute angle between the lines $\frac{x-1}{1} = \frac{y-2}{-1} = \frac{z-3}{2}$ and $\frac{x-1}{2} = \frac{y-2}{1} = \frac{z-3}{1}$ is _____.
(a) 60° (b) 30° (c) 45° (d) 90° (2)
- iv. The vector equation of the plane passing through point A ($\vec{a}$) and parallel to $\vec{b}$ and $\vec{c}$ is _____.
(a) $(\vec{r} - \vec{a}) \times (\vec{b} \times \vec{c}) = \vec{0}$ (b) $(\vec{r} - \vec{a}) \cdot (\vec{b} + \vec{c}) = 0$
(c) $(\vec{r} - \vec{a}) \cdot (\vec{b} \times \vec{c}) = 0$ (d) $(\vec{r} - \vec{a}) \times (\vec{b} - \vec{c}) = \vec{0}$ (2)
- v. If $x = at^4$, $y = 2at^2$, then $\frac{dy}{dx} =$ _____.
(a) $\frac{1}{t^2}$ (b) t^2 (c) $2t^2$ (d) $-\frac{1}{t^2}$ (2)
- vi. The area bounded by the curve $y = 2x$, the Y-axis, the X-axis and $x = 3$ is _____.
(a) 3 sq. units (b) 6 sq. units (c) 9 sq. units (d) 12 sq. units (2)





- vii. The differential equation $y \frac{dy}{dx} + x = 0$ represents family of _____.
 (a) circle (b) parabola (c) ellipse (d) hyperbola (2)
- viii. If $f(x) = kx^2(1-x)$, for $0 < x < 1$
 $= 0$, otherwise, is the probability distribution function of a random variable X, then
 the value of k is _____.
 (a) 12 (b) 10 (c) -9 (d) -12 (2)

Q.2. Answer the following questions:**[4]**

- i. Write the domain of inverse secant function. (1)
- ii. Find the value of k, if lines represented by $kx^2 + 4xy - 4y^2 = 0$ are perpendicular to each other. (1)
- iii. Evaluate: $\int \frac{1}{x\sqrt{\log x}} dx$ (1)
- iv. Obtain the differential equation by eliminating the arbitrary constant from the equation $y^2 = 4ax$. (1)

SECTION – B**Attempt any EIGHT of the following questions:****[16]****Q.3. Write the following compound statements symbolically:**

- (i) Nagpur is in Maharashtra and Chennai is in Tamilnadu. (2)
- (ii) If ΔABC is right angled at B, then $m\angle A + m\angle C = 90^\circ$ (2)

Q.4. Find the co-factors of the elements a_{11} and a_{21} of matrix $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$. (2)**Q.5. Find the solution of $\cos \theta = \frac{1}{2}$, where $0 \leq \theta < 2\pi$.** (2)**Q.6. Find the joint equation of lines passing through the origin having slopes, 2 and 3.** (2)**Q.7. If the vectors $-3\hat{i} + 4\hat{j} - 2\hat{k}$, $\hat{i} + 2\hat{k}$ and $\hat{i} - p\hat{j}$ are coplanar, then find the value of p.** (2)**Q.8. Find the direction cosines of the vector $2\hat{i} + 2\hat{j} - \hat{k}$.** (2)**Q.9. Find the equation of tangent to the curve $y = x^2 + 2e^x + 2$ at the point (0, 4).** (2)**Q.10. Evaluate: $\int \frac{3^x - 4^x}{5^x} dx$** (2)**Q.11. Evaluate: $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \cos x \, dx$** (2)**Q.12. Find the area of the region bounded by the curve $y = x^2$, the X-axis and the lines $x = 1$, $x = 3$.** (2)**Q.13. Find the integrating factor (I. F.) of the differential equation:**

$$\frac{dy}{dx} + y = e^{-x} \quad (2)$$

Q.14. Given that $X \sim B(n, p)$.

If $p = 0.6$ and $E(X) = 6$, find n and $\text{Var}(x)$. (2)

SECTION – C**Attempt any EIGHT of the following questions:****[24]****Q.15. Prove that $\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3 = \pi$** (3)**Q.16. If one of the lines given by $ax^2 + 2hxy + by^2 = 0$ bisects an angle between the co-ordinate axes then show that $(a+b)^2 = 4h^2$** (3)



- Q.17.** Let $A(\vec{a})$ and $B(\vec{b})$ be any two points in the space and $R(\vec{r})$ be the third point on the line AB dividing the segment AB externally in the ratio $m : n$. Then prove that $\vec{r} = \frac{m\vec{b} - n\vec{a}}{m - n}$ (3)
- Q.18.** Find the position vector of point P such that OP is inclined to X – axis at 45° and to Y – axis at 60° and $OP = 12$ units. (3)
- Q.19.** Find the vector equation of the line passing through the point having position vector $2\hat{i} + \hat{j} - 3\hat{k}$ and perpendicular to vectors $\hat{i} + \hat{j} + \hat{k}$ and $\hat{i} + 2\hat{j} - \hat{k}$. (3)
- Q.20.** The foot of perpendicular drawn from the origin to a plane is $M(1, 2, 0)$. Find the vector equation of the plane. (3)
- Q.21.** If $\log_{10} \left(\frac{x^3 - y^3}{x^3 + y^3} \right) = 2$, show that $\frac{dy}{dx} = \frac{-99x^2}{101y^2}$ (3)
- Q.22.** A wire of length 36 meters is bent in the form of rectangle. Find its dimensions if the area of the rectangle is maximum. (3)
- Q.23.** Evaluate: $\int \frac{e^x}{1 + e^{-x}} dx$ (3)
- Q.24.** Solve the differential equation:
 $\frac{dy}{dx} = (4x + y + 1)^2$ (3)
- Q.25.** The probability distribution of X is as follows:
- | | | | | | |
|----------|-----|---|----|----|---|
| X | 0 | 1 | 2 | 3 | 4 |
| P(X = x) | 0.1 | K | 2K | 2K | K |
- Find: (i) K (ii) $P(X < 2)$ (iii) $P(X \geq 3)$ (3)
- Q.26.** Ten eggs are drawn successively with replacement from a lot containing 10% defective eggs. Find the probability that there is at least one defective egg. (3)

SECTION – D

Attempt any FIVE of the following questions:

[20]

- Q.27.** Construct the truth table for the statement pattern $(p \rightarrow q) \wedge [(q \rightarrow r) \rightarrow (p \rightarrow r)]$ and interpret your result. (4)
- Q.28.** Solve the following equations by using method of inversion:
 $x - y + z = 4, 2x + y - 3z = 0, x + y + z = 2$ (4)
- Q.29.** Prove that, in $\triangle ABC$,
 $\tan\left(\frac{A - B}{2}\right) = \left(\frac{a - b}{a + b}\right) \cot\left(\frac{C}{2}\right)$ (4)
- Q.30.** Solve the linear programming problem by graphical method:
 Maximize: $z = 3x + 5y$
 Subject to: $x + 4y \leq 24$
 $3x + y \leq 21$
 $x + y \leq 9$ and $x \geq 0, y \geq 0$
 Also find maximum value of z. (4)





Q.31. If $y = f(x)$ is a differentiable function of x on an interval I and y is one-one, onto and $\frac{dy}{dx} \neq 0$ on I ,

then prove that $\frac{dx}{dy} = \frac{1}{\frac{dy}{dx}}$.

where $\frac{dy}{dx} \neq 0$. Hence prove that $\frac{d}{dx}(\cot^{-1}x) = \frac{-1}{1+x^2}$ (4)

Q.32. The volume of the spherical ball is increasing at the rate of 4π cc/sec. Find the rate at which the radius and the surface area are changing when the volume is 288π cc. (4)

Q.33. Prove that: $\int \frac{1}{x^2 - a^2} dx = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + c$

Hence evaluate: $\int \frac{1}{x^2 - 3} dx$ (4)

Q.34. Evaluate: $\int_0^{\frac{\pi}{2}} \cos^3 x dx$ (4)

